98. The fraction of a radioactive material which remains active after time t is $9/16$. The fraction which remains active after time $t/2$ will be

A. $4/5$ B. $7/8$ C. $3/5$ D. $3/4$

Solution: $\frac{9}{16} = \left(\frac{1}{2}\right)^{t/T}$ and $\frac{N}{No} = \left(\frac{1}{2}\right)^{\frac{t}{2T}}$

$\left(\frac{N}{N_o}\right)^2 = \left(\frac{1}{2}\right)^{t/T}$ or $\left(\frac{N}{No}\right)^2 = 9/16 \Rightarrow \frac{N}{No} = 3/4$ **Answer: D**

99. Half – life of a radioactive substance is 20 minutes. The time between 20% and 80% decay will be

A. 40 minutes B. 20 minutes
C. 25 minutes D. 30 minutes

Solution: $\frac{20}{80} = \frac{1}{2^{1/20}}$ or $\frac{1}{4} = \frac{1}{2^{1/20}}$ or $\frac{1}{2^2} = \frac{1}{2^{1/20}}$

$\Rightarrow \frac{t}{20} = 2$ or $t = 40$ minutes **Answer: A**

100. The percentage of quantity of a radioactive material that remains after 5 half-lives will be

A. 31% B. 3.125%
C. 0.35% D. 1 %

Explanation:

$\frac{N}{N_o} = \frac{1}{2^{\frac{5T}{T}}} = \frac{1}{2^5} = \frac{1}{32} = 3.125\%$ **Answer: B**

101. A radioactive nucleus undergoes a series of decays according to the scheme

$$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$$

If the mass number and atomic number of A are 180 and 72 respectively, then what are these number for A_4?

A. 172 and 69 B. 174 and 70
C. 176 and 69 D. 176 and 70

Explanation: Two α – particles reduce mass number by 8. New mass number = 180 – 8 = 172. And reduce charge number by 4. Emission of β – particle increases charge number by 1. New charge number = 72 – 4 + 1 = 69 Hence, 172 and 69. **Answer: A**

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102. The sodium nucleus $^{23}_{11}Na$ contains

A. 11 electrons B. 23 protons
C. 12 protons D. 12 nutrons

Answer: D

103. Which of the following cannot be emitded by radioactive substance during their decay?

A. electrons B. protons
C. neutrons D. helium nuclei

Answer: B

104. During a nuclear fusion reaction

A. A heavy nucleus breaks into two fragments by its self
B. A light nucleus bombarded by thermal neutrons breaks up
C. A heavy nucleus bombarded by thermal neutrons breaks up
D. Two light nuclei combine to give a heavier nucleus and possible other products

Answer: D

105. The mass number of a nucleus is

A. Always less then its atomic number
B. Always more than its atomic number
C. Sometimes equal to its atomic number
D. Sometimes more than and sometimes equal to its atomic number.

Answer: D

106. The equation $4\left(^{1}_{1}H^{+}\right) \rightarrow {}^{2}_{4}H\,e^{++} + 2e^{-} + 26$ Mev represents

A. β – decay B. γ – dacay
C. fusion D. fission**Answer: C**

107. As a result of radioactive decay a $_{92}U^{238}$ nucleus is changed to a $_{91}P_a{}^{234}$ nucleus. During this decay the particles remitted are

A. one proton and two neutrons
B. one $\propto$ - particle and one β – particle
C. two β – particles and one neutron
D. two β – particles and one proton

Explanation: $^{238}_{92}U \xrightarrow{\alpha} {}^{234}_{90}Th \xrightarrow{\beta^-} {}^{234}_{91}Pa$ **Answer: B**

108. In a given reaction

$$_{Z}X^{A} \longrightarrow_{z+1} Y^{A} \longrightarrow_{z-1} K^{A-4} \longrightarrow_{z-1} K^{A-4}$$

Radioactive radiations are emitted in the sequence

A. α, β, γ B. β, α, γ
C. γ, α, β D. β, γ, α **Answer: B**

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109. The fusion of hydrogen into helium is more likely to take place.
A. At high temperature and high pressure
B. At high temperature and low pressure
C. At low temperature and low pressure
D. At low temperature and high pressure

Answer: A

110. Consider α, β and Υ rays, each having an energy of 0.5 Mev. In increasing order of penetrating powers, the radiations are
A. α, β, γ B. α, γ, β
C. β, γ, α D. γ, β, α

Answer: A

111. In the reaction
$_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + x$, x is
A. electron B. proton
C. photon D. neutron

Answer: C

112. The main source of solar energy is
A. nuclear fission
B. gravitational contraction
C. nuclear fussion
D. combustion of coal and hydrogen

Answer: C

113. When ^7_3Li nuclei are bombarded by protons and the resultant nuclei are ^8_4Be, the emitted particles will be
A. α – particles B. γ – photons
C. β – particles D. neutrons

Explanation: $^7_3Li + {}^1_1H \longrightarrow {}^8_4Be + \gamma$

Answer: C

Unit 16: Conceptual Questions

1. Is light a wave or a particle?
2. Which has more energy; a photon of ultraviolet radiation or a photon of yellow light?
3. Are black bodies black? Explain
4. All objects radiate energy. Why then are we not able to see objects in a dark room?
5. What is photon? Explain

Unit 16: Problems

Photoelectric Effect

6. The energy of photon varies directly with its;
A. frequency B. Wavelength C. speed D. Rest mass

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7. Which color of light has the greatest energy per photon?
   A. Red B. Green C. Blue D. Violet
8. Light demonstrates the characteristics of:
   A. particle only C. Both particle and wave
   B. wave only D. neither particle nor wave
9. Which phenomenon best supports the particle theory of light?
   A. photoelectric effect B. diffraction
   C. interference D. polarization

Radioactivity and Nuclear Reaction

10. The sun's energy is believed to be a result of:
    A. Nuclear fission B. oxidation of hydrogen
    C. oxidation of helium D. Nuclear fusion
11. The half-life of uranium is 8.5 days. What fraction of the sample remains un decayed after 34 days
    A. $1/2$ B. $1/4$ C. $1/8$ D. $1/16$
12. The half-life of radium −224 is 3.5 days. What fraction of the sample remains undecayed after fourteen days? (If needed use) $\ln 2 = -\ln\left(\frac{1}{2}\right) \simeq 0.693$ [UEE: 2006]
    A. $1/2$ B. $1/4$ C. $1/8$ D. $1/16$

Unit 16: Explanation to Questions

1. Light is both particle and wave – dual nature
2. E = hf. The frequency of uv > The frequency of yellow light. Thus, UV light has more energy than yellow light
3. No. The black appearance, is due to the fact that our eye is sensitive only to visible light range.
4. All objects do radiate energy, but at room temperature, this energy is primarily in the infrared region of the electromagnetic radiation spectrum, which our eyes cannot defect.
5. Photon is a quantum (packet) of light

Unit 16: Solution to Problems

6. A 7. D 8. C 9. A 10. D

11. $n = \frac{34}{8.5} = 4, \quad \frac{N}{N_o} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$ **Answer: D**

12. $n = \frac{14}{3.5} = 4, \quad \frac{N}{N_o} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$ **Answer: D**

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